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Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

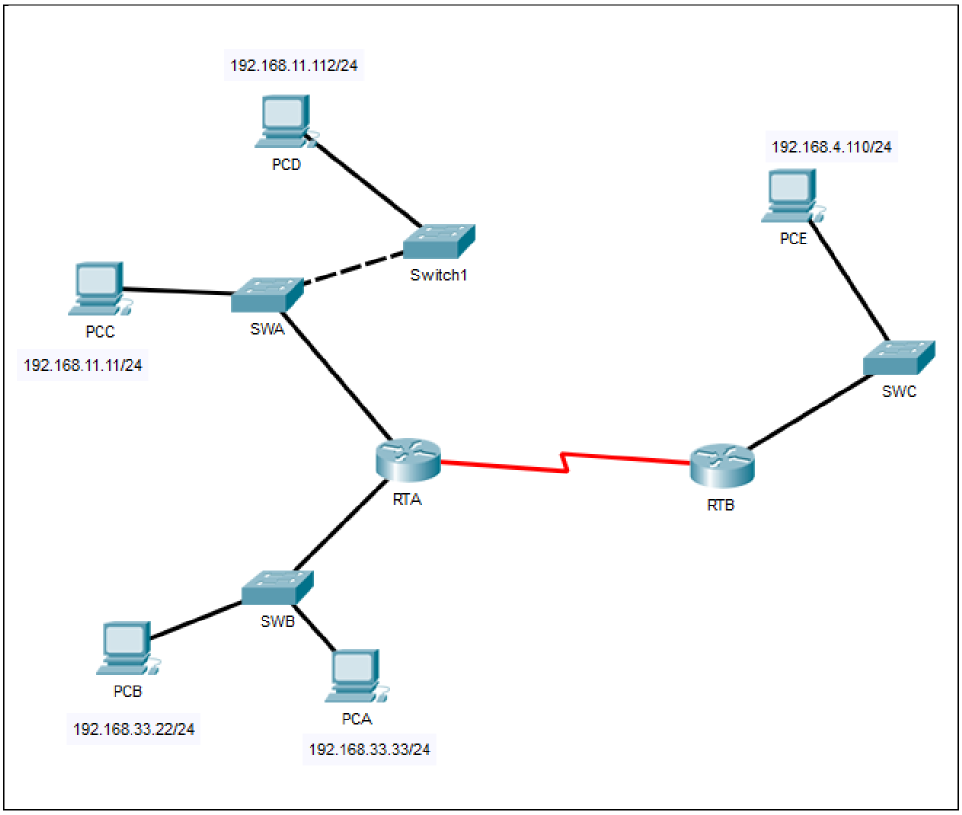


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.11.1            -   0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1            -   000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.4.1             -   0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

SWA

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4a00.0e91    DYNAMIC     Fa0/1
   1    000c.8546.7d85    DYNAMIC     Fa1/1
SWA#
```

SWB

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf0c.593a    DYNAMIC     Fa0/1
SWB#
```

SWC

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.977a.b614    DYNAMIC     Fa0/1
SWC#
```

d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **`arp -a`** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCB

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

e. What are your thoughts on the results?

Ans: There are no entry for all PCs because all the PCs didn’t receive any ARP request so there will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

a. Click **PCA** and click the Desktop tab > Command Prompt.
b. Enter the **ping 192.168.33.22** command. This may take a few seconds.
c. In the Command prompt of PCA, type **arp –a**. Paste the result of this command here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.22         0060.47ea.a746        dynamic

C:\>
```

d. In the Command prompt of PCB, type **arp –a**. Paste the result of this command here

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.33         0002.1755.9a06        dynamic

C:\>
```

e. In the Command prompt of PCC, PCD abd PCE, type **arp –a**. Paste the result of this command here.

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

a. Click **PCC** and click the Desktop tab > Command Prompt.

b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp –a**. Paste the result after these commands here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 5ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic

C:\>
```

c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
  192.168.11.112        0001.6462.0278        dynamic

C:\>
```

d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 10ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
  192.168.11.112        0001.6462.0278        dynamic

C:\>
```

e. Discuss the results you got from all the commands on PCC.

Ans: The entries of ARP are shown in the ARP table only after PCC pings PCB, PCC and PCE. There are no ARP entries are found before these ping activities. The ARP table displays the entries once PCB, PCD and PCE receive the ARP request from PCC and subsequently receive pings from PCB< PCD and PCE.

f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.11.1            -   0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11           6   00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1            -   000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22           6   0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

g. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.4.1             -   0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110           6   0060.702D.7C08  ARPA   FastEthernet0/0
RTB#
```

Step 3: Switch MAC address table.

a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.6462.0278    DYNAMIC     Fa1/1
   1    0002.4a00.0e91    DYNAMIC     Fa0/1
   1    000c.8546.7d85    DYNAMIC     Fa1/1
   1    00d0.d39a.c0d9    DYNAMIC     Fa2/1
SWA#
```

b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf0c.593a    DYNAMIC     Fa0/1
SWB#
```

c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.977a.b614    DYNAMIC     Fa0/1
   1    0060.702d.7c08    DYNAMIC     Fa1/1
SWC#
```

d. Do switches use arp table? (Y/N)

   Ans: Yes

e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

   Ans: Layer 3 switches use an Address Resolution Protocol (ARP) table to map IP addresses to MAC addresses. ARP tables are used to forward packets or route them based on a routing protocol.

f. What information does the command ***show mac-address-table*** gives?

   Ans: To view the MAC address table entries on a switch, the term 'DYNAMIC' specifically pertains to display MAC address that have been learned by the switch during its operation.

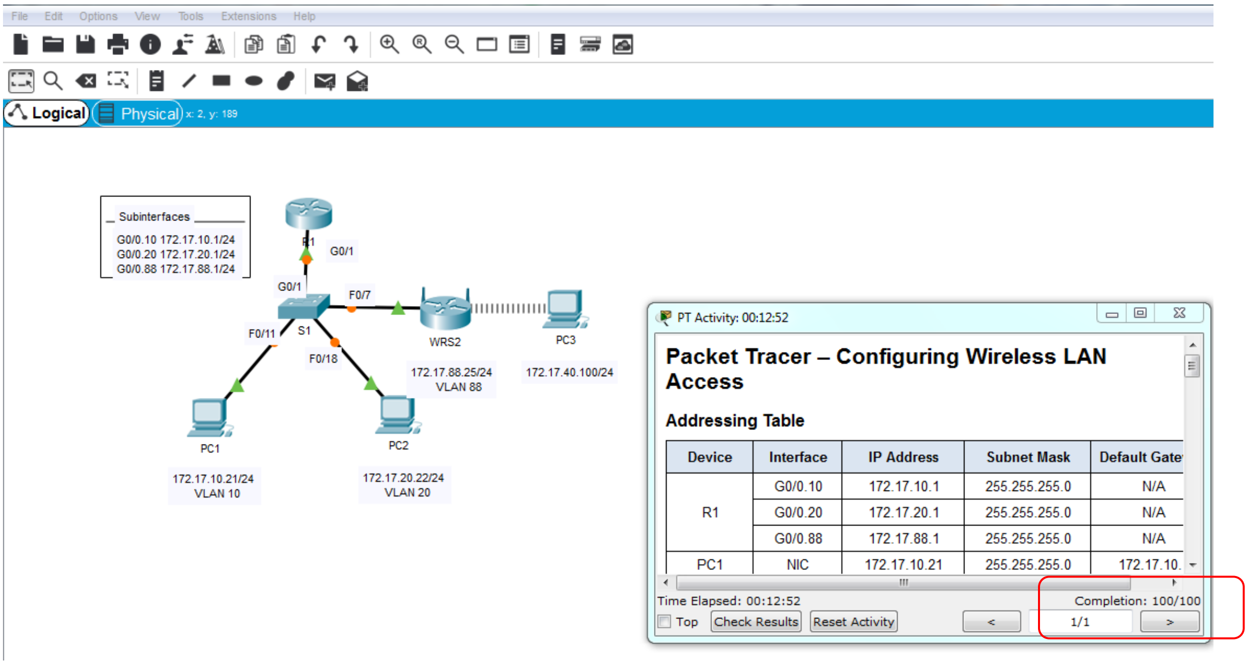
Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

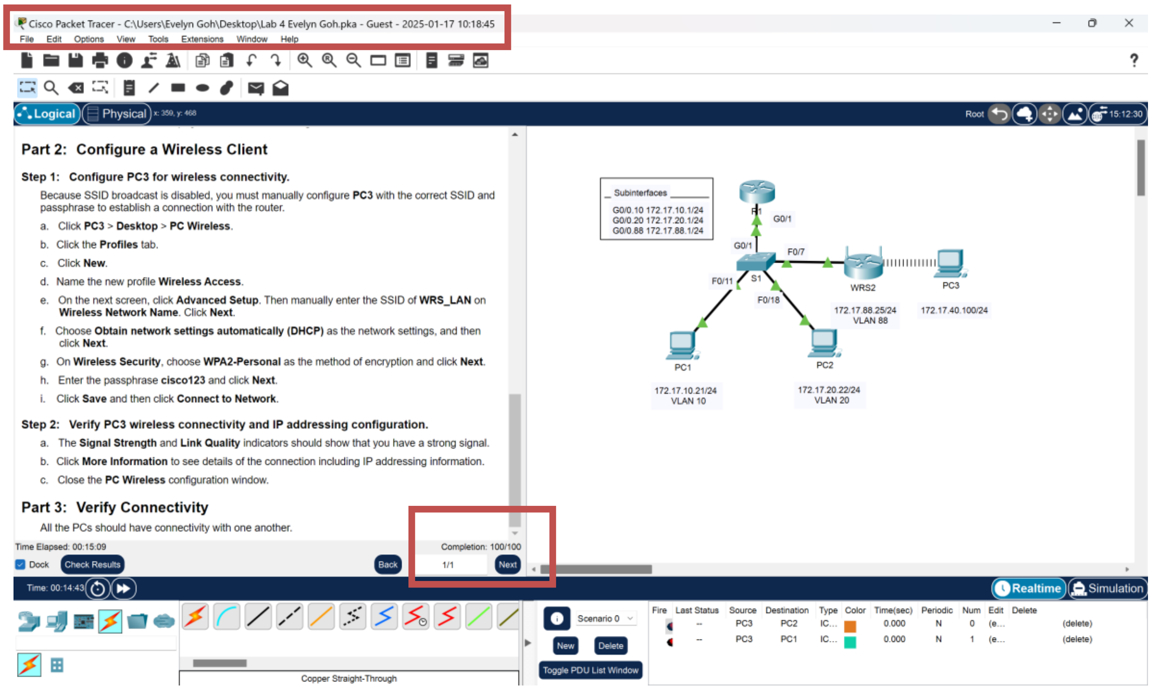
Step1: Change the filename of Lab 4.

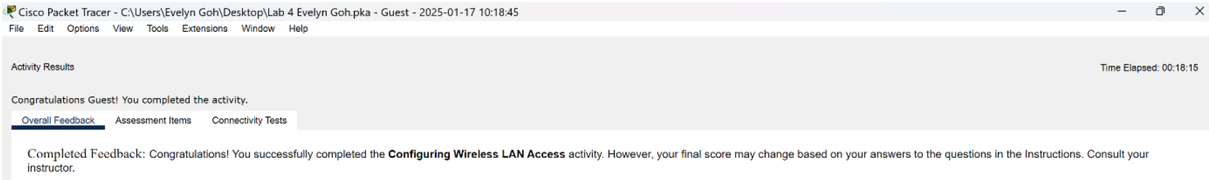
a. Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*

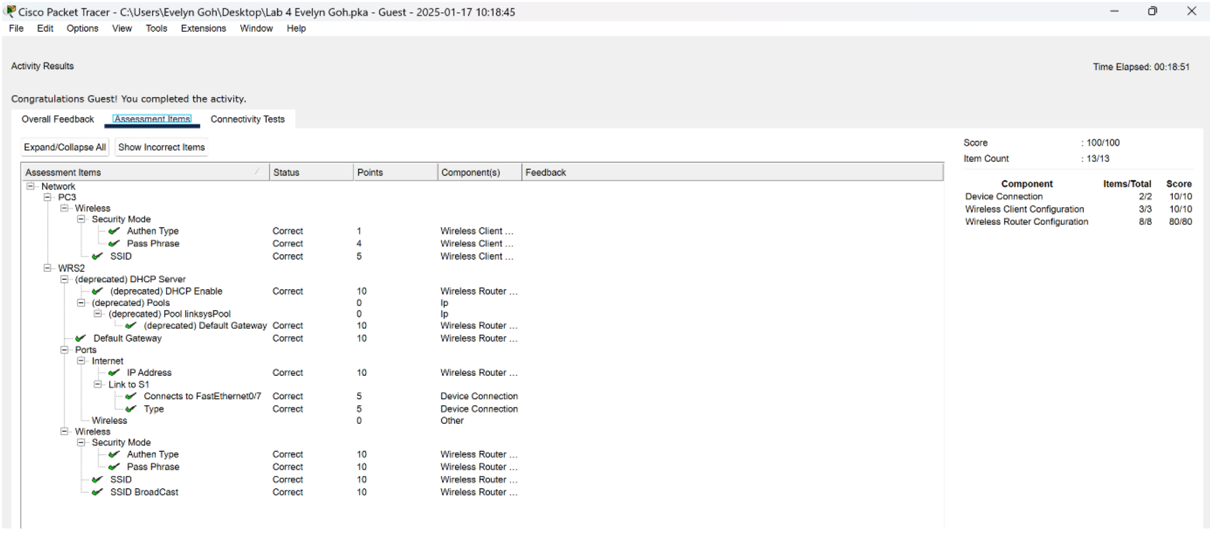
b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.



c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.




Cisco Packet Tracer - C:\Users\Evelyn Goh\Desktop\Lab 4 Evelyn Goh.pka - Guest - 2025-01-17 10:18:45
File Edit Options View Tools Extensions Window Help
Activity Results
Time Elapsed: 00:18:15
Congratulations Guest! You completed the activity.
Overall Feedback
Assessment Items
Connectivity Tests
Completed Feedback: Congratulations! You successfully completed the Configuring Wireless LAN Access activity. However, your final score may change based on your answers to the questions in the Instructions. Consult your instructor.


Cisco Packet Tracer - C:\Users\Evelyn Goh\Desktop\Lab 4 Evelyn Goh.pka - Guest - 2025-01-17 10:18:45
File Edit Options View Tools Extensions Window Help
Activity Results
Time Elapsed: 00:18:51
Congratulations Guest! You completed the activity.
Overall Feedback
Assessment Items
Connectivity Tests
Expand/Collapse All
Show Incorrect Items
Assessment Items
Status
Points
Component(s)
Feedback
Network
PC3
Wireless
Security Mode
Authen Type Correct 1 Wireless Client ...
Pass Phrase Correct 4 Wireless Client ...
SSID Correct 5 Wireless Client ...
WRS2
(deprecated) DHCP Server
(deprecated) DHCP Enable Correct 10 Wireless Router ...
(deprecated) Pools 0 Ip
(deprecated) Pool linksysPool 0 Ip
(deprecated) Default Gateway Correct 10 Wireless Router ...
Default Gateway Correct 10 Wireless Router ...
Ports
Internet
IP Address Correct 10 Wireless Router ...
Link to S1
Connects to FastEthernet0/7 Correct 5 Device Connection
Type Correct 5 Device Connection
Wireless 0 Other
Wireless
Security Mode
Authen Type Correct 10 Wireless Router ...
Pass Phrase Correct 10 Wireless Router ...
SSID Correct 10 Wireless Router ...
SSID BroadCast Correct 10 Wireless Router ...
Score : 100/100
Item Count : 13/13
Component Items/Total Score
Device Connection 2/2 10/10
Wireless Client Configuration 3/3 10/10
Wireless Router Configuration 8/8 80/80